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An Impossibility Result for ETF Target Geometry, and an Equinorm Remedy for Imbalanced Analytic Continual Learning

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Paper & Code

TL;DR Neural-Collapse **ETF targets** in exact **analytic continual learning** are *provably identical* to one-hot targets, and the literal "progressively expand the ETF" recipe causes catastrophic forgetting. What helps is **equal weight norms**: post-hoc **equinorm** gives **+32 tail / +16 average** accuracy points at 100 : 1 imbalance.

1 Can ETF targets help analytic CL?

- **Analytic CL** (ACIL [1], GACL [2]): frozen backbone + ridge head $W = (\Phi^\top\Phi + \lambda I)^{-1}\Phi^\top Y$, updated exactly per task by Sherman–Morrison–Woodbury. Exemplar-free.
- **ProNC** [3]: SGD-trained CL improves with a progressively expanding **Simplex ETF** target.
- **Simplex ETF** [4] $A = [a_1, \ldots, a_K]$: $\|a_c\| = 1$ *(equinorm)*, $\langle a_c, a_{c'}\rangle = -\frac{1}{K-1}$ *(equiangular)*.

Q: swap ACIL's one-hot targets for an ETF? A: it changes nothing.

2 Theorem: target geometry is invisible

Targets $E = YA^\top$, scores $s_A(x) = \phi^\top W_A A$. For any fit linear in its targets (ridge, weighted ridge, ACIL/GACL, RanPAC [5]):

$$s_A(x) = s_{\text{onehot}}(x)\, A^\top A, \qquad A^\top A \overset{\text{ETF}}{=} \tfrac{K}{K-1} I - \tfrac{1}{K-1}\mathbf{1}\mathbf{1}^\top$$

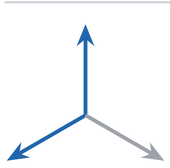
$$\Rightarrow\ s_A(x)_c = \underbrace{\tfrac{K}{K-1}}_{>0}\, s_{\text{onehot}}(x)_c \;-\; \underbrace{\tfrac{1}{K-1}\textstyle\sum_{c'} s_{\text{onehot}}(x)_{c'}}_{\text{same for every class}}$$

Identical argmax for every input, every $m \geq K-1$, every λ, every task, any sample weighting.

Verified on real features: score gap $< 1.1 \times 10^{-14}$.

Escapes: **compressed frames** $m < K-1$ (no ETF exists; costly, $\leq 7.8\%$ accuracy vs. 68.6%) or **non-linear** post-processing of W (**4**).

3 Simplex ETFs are not nested



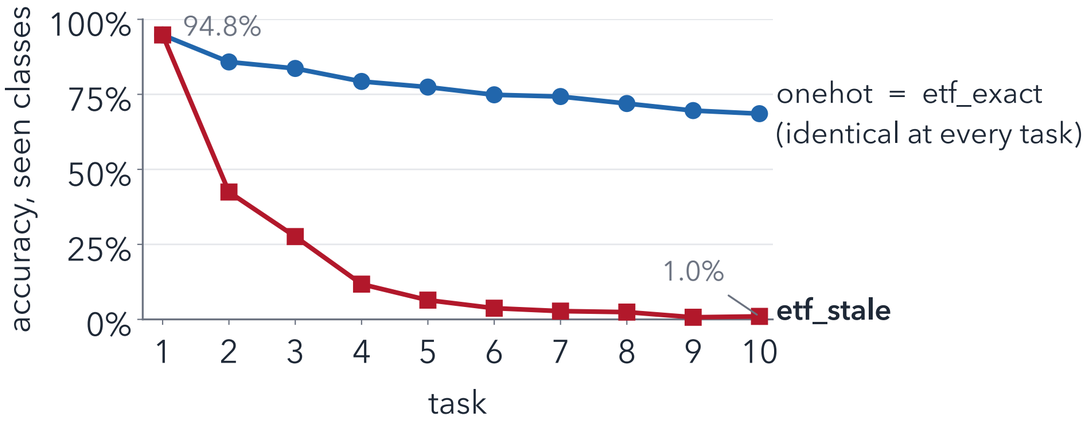
$K = 3$
$\cos = -\frac{1}{2}$



$K = 4$
$\cos = -\frac{1}{3}$

Proposition. ETFs for K and $K+1$ classes share **at most one** vertex: any fixed pair's cosine would have to be both $-\frac{1}{K-1}$ and $-\frac{1}{K}$.

Fused methods must recompute the ETF each task (`etf_exact` ≡ one-hot) or keep old target blocks in their old frame (`etf_stale`):



CIFAR-100, 10 tasks, ViT-B/16. Staleness alone brings back catastrophic forgetting ($4\times$ that of `onehot`); same on TinyImageNet and ResNet-50.

4 Split the ETF, apply it to W

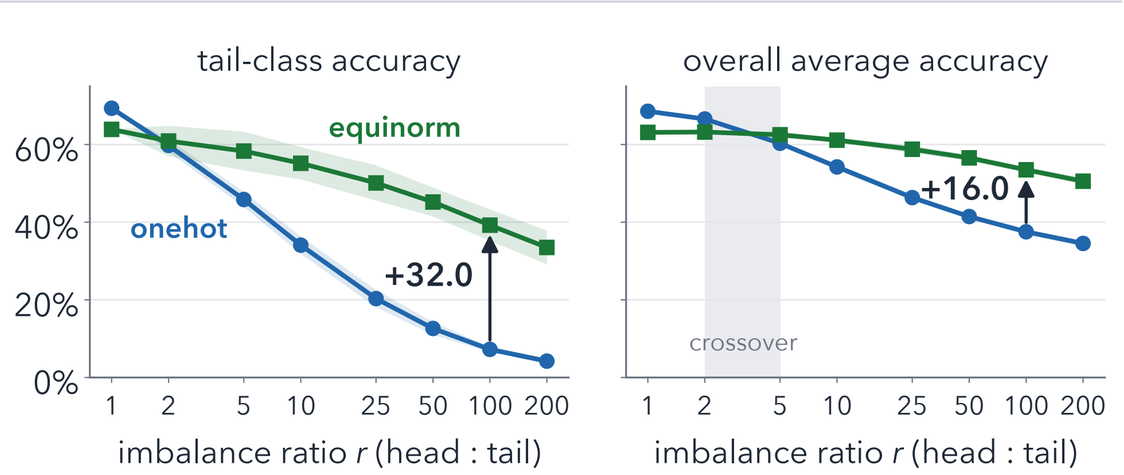
Targets are inert, so impose each ETF property on the *fitted weights*, a non-linear map outside the theorem:

- **equinorm**: equalize column norms of W, keep its angles. (τ-normalization [6] with τ=1; new here: in analytic CL, split from equiangularity.)
- **equiangular**: set all pairwise cosines to $-\frac{1}{K-1}$, keep its norms.

Why norms? From $r = 1$ to 100, the norm ratio $\|w\|_{\max}/\|w\|_{\min}$ of the fitted W grows **5.2** → **43.7**; the spread of its cosines barely moves (0.72 → 0.78). Imbalance lives in the norms.

Setup. Frozen ViT-B/16 and ResNet-50; CIFAR-100 and TinyImageNet, 10 tasks. Long tail: exponential per-class budget, head : tail ratio r. 5 seeds; all heads fit in lockstep on identical features.

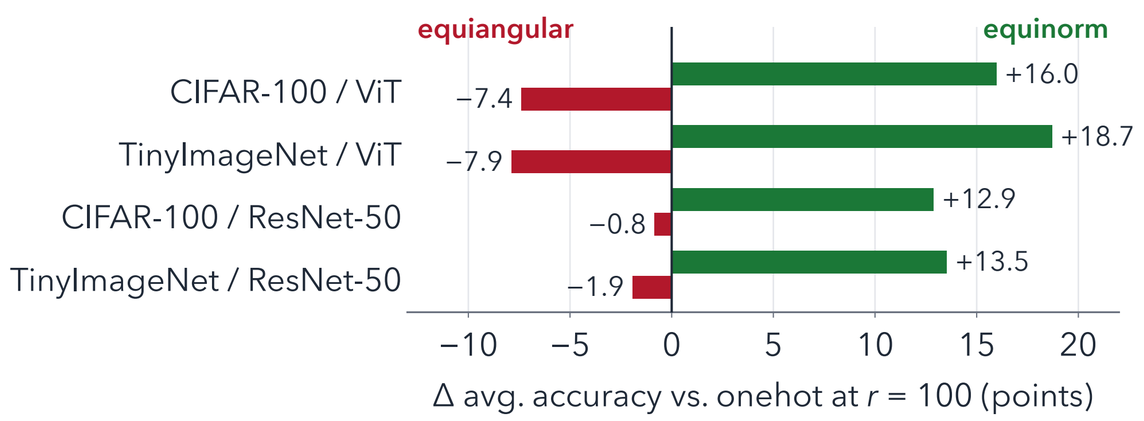
5 Equinorm rescues the long tail



tail classes	all classes	head classes
7.2% → **39.2%**	37.5% → **53.5%**	67.8% → **67.8%**
+32.0	**+16.0**	**≈ 0**

CIFAR-100/ViT, mean ± s.d. over 5 seeds; tiles at $r = 100$ (onehot → equinorm). Cost near balance: -5.4 points at $r = 1$, -3.4 at $r = 2$.

6 Equinorm transfers; equiangularity hurts



Equinorm helps in all four backbone × dataset cells; equiangular falls below `onehot` in all four.

Take-home

1. ETF targets in analytic CL are **provably identical** to one-hot targets.
2. The literal progressive-ETF recipe leaves **stale targets**, which bring back catastrophic forgetting.
3. The useful half of Neural-Collapse geometry is **equinorm on the weights**; equiangularity is at best inert, usually harmful.

References [1] H. Zhuang et al. ACIL: Analytic class-incremental learning with absolute memorization and privacy protection. NeurIPS 2022. [2] H. Zhuang et al. GACL: Exemplar-free generalized analytic continual learning. NeurIPS 2024. [3] Z. Wang, W. Yu, L. Yang, S. Lin. Rethinking continual learning with progressive neural collapse. ICLR 2026. [4] V. Papyan, X. Y. Han, D. L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. PNAS 2020. [5] M. D. McDonnell et al. RanPAC: Random projections and pre-trained models for continual learning. NeurIPS 2023. [6] B. Kang et al. Decoupling representation and classifier for long-tailed recognition. ICLR 2020.

Code github.com/berksezer9/etf-targets-acil